

Erdős Problem #626

EP-626

L1 - TRACTABLE

PARTIALLY SOLVED

Graph Theory

ORIGINAL PROBLEM STATEMENT

Let $k \geq 4$ and $g_k(n)$ denote the largest m such that there is a graph on n vertices with chromatic number k and girth $> m$ (i.e. contains no cycle of length $\leq m$). Does $\lim_{n \rightarrow \infty} \frac{g_k(n)}{\log n}$ exist?

Conversely, if $h^{(m)}(n)$ is the maximal chromatic number of a graph on n vertices with girth $> m$ then does $\lim_{n \rightarrow \infty} \frac{\log h^{(m)}(n)}{\log n}$ exist, and what is its value?

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Archive: <https://www.unsolvedmath.com/problems?difficulty=1>

Dataset: <https://huggingface.co/datasets/ulamai/UnsolvedMath>